Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/27/2017
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center)		
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	NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A. Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat		

Task Description:	thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder and wake regions result in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with several imbedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	1. Numerical modeling for Burning and Suppression of Solids (BASS) experiments 1.1 Flat SIBAL samples: Flame growth, spread, and quenching extinction over a thin composite cotton-fiberglass fabric blend (referred to as the SIBAL fabric) were studied in low-speed concurrent purely forced flows aboard the International Space Station. The tests were conducted in a small flow duct within the Microgravity Science Glovebox. The fuel samples measured 1.2 and 2.2 cm wide and 10 cm long. Ambient oxygen was varied from 21% down to 16% molar concentration and flow speed from 55 cm/s down to 1 cm/s. A slow purely forced flow resulted in a small flame, enabling us to observe the entire history of flame development including ignition, flame growth, steady spread (in some cases), and decay at the end of the sample. In addition, by decreasing flow velocity during some of the tests, low-speed flame quenching extinction limits were determined as a function of oxygen percentage. The quenching speeds were found to be between 1 and 5 cm/s with higher extinction speeds in lower oxygen atmospheres. The shape of the quenching boundary supports the prediction by earlier theoretical models. These long duration microgravity experiments provide a rare opportunity for solid fuel combustion since microgravity time in ground-based facilities is generally not sufficient. This is the first time that a low speed quenching boundary in concurrent spread is mapped in a clean and unambiguous manner. A previously developed three-dimensional transient model is modified to compare with the experiment. The modification includes the use of two-step SIBAL fabric pyrolysis kinetics where the rate constants are determined using Thermo-Gravimetric Analysis data. The model yields good quantitative comparison on the quenching boundary, the flame transient development, and the steady flame spread rates. 1.2 Radiation Model and Spherical PMMA samples: Gas phase radiative heat transfer becomes especially important for flame spread over solid fuel in microgravity near the low velocity quenching limit, also known as radiative quenching limit. Near this limit the flame color of PMMA is blue due to the low soot content and therefore, spectral dependency of the radiative heat transfer becomes important. Commonly used radiation treatments such as gray gas model and wide band model face difficulty in obtaining accurate results in such condition. Moreover, being a diffusion flame prevents the usage of some of the popular special approximations such as spherical harmonics method since the absorption coefficient becomes zero in some wave lengths. This research overcomes such challenges by solving the radiation field around a PMMA sphere by Statistical Narrow-Band Correlated-k (SNB-CK) spectral model combined with three dimensional discrete ordinate method. The model is applied to an experimental configuration from the recent microgravity BASS project aboard the International Space Station. A PMMA sphere (initial diameter of 2 cm) is placed in a small wind tunnel (8 cm x 8 cm x 17.5 cm) within the microgravity glovebox where flow speed and oxygen percentage are varied. The recorded flame spread rate after ignition at the forward stagnation point and radiative quenching limiting flame when flow velocity is reduced are compared with the model results. 2. Ground based Experiments 2.1 Droptower tests of instrumented PMMA sphere samples: A technique of making cast PMMA sphere sample with 5 small thermocouples embedded at different locations has been devised and three proto samples have been made by an outside manufacturer. One of the most recently made samples has been tested in the 5-s droptower. This is a transient experiment where a sample is ignited in normal gravity, left burning for a period of time until the prescribed sub-surface temperature gradient is reached before dropped in microgravity. Transient flame development is recorded. Three tests have been performed in 17% O2 with a forced 2 to 6 cm/s flow. Flames go out during the drop in all the three cases. To determine the flammable limits, more tests will be conducted once additional fabricated samples arrive. One of the purposes of this series of tests is to help us to define the preheating levels to be used in the flight experiment. 2.2. Gravimetric Measurement of Solid and Liquid Fuel Burning Rate near and at the Low Oxygen Extinction Limit: A precision mass balance coupled with a variable oxygen flow tunnel/nozzle enables us to measure the burning rate of solid and liquid fuels as a function of ambient oxygen percentage all the way to the extinction limit. Two sample configurations have been studied. The first is a liquid-fueled wick flame (ethanol tea lamp). The total burning rates (mass/time) are measured as a function of wick length and oxygen percentage. Near the low oxygen limit, limit-cycle flame oscillation has been found that can last for many minutes to an hour. The averaged mass burning rate of the oscillatory flame is about one-half of that of the neighboring steady flame. In the second configuration, local burning rates (mass/area time) of two Polymethylmethacrylate spherical samples have been measured in the flame stabilization zone. Each sample has a different amount of heat loss and a different oxygen limit. The critical burning rate at their respective oxygen limit are different. This implies that critical burning rate is not a property of the material alone and it

	should not be used as the only criterion to judge the extinction or the ignition of materials.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Zhao X, Liao Y-TT, Johnston MC, T'ien JS, Ferkul PV, Olson SL. "Concurrent flame growth, spread, and quenching over composite fabric samples in low speed purely forced flow in microgravity." Proceedings of the Combustion Institute. 2017;36(2):2971-8. http://dx.doi.org/10.1016/j.proci.2016.06.028 , Jan-2017
Dissertations and Theses	Shang W. "Laboratory-scale evaluation of aramid blend material as a fire barrier for flexible polyurethane foam." MS thesis, Case Western Reserve University, 2016. , Jun-2016
Papers from Meeting Proceedings	Shang,W, T'ien JS. "Laboratory evaluation of Nomex(R) IIIA aramid fabric as a flammability barrier material in a variable oxygen and pressure atmosphere." 46th International Conference on Environmental Systems, Vienna, Austria, July 10-14, 2016. 46th International Conference on Environmental Systems, Vienna, Austria, July 10-14, 2016. ICES paper ICES-2016-38. , Jul-2016